

VINAY SAMUEL

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EDUCATION

University of Maryland, College Park	2025–2026
Master of Science in Computer Science	GPA: 4.00
Carnegie Mellon University	2021–2025
Bachelor of Science in Statistics and Machine Learning	

WORK EXPERIENCE

Scale AI	San Francisco, CA
<i>Machine Learning Research Intern</i>	May 2026–Present
- Led first author research on task agnostic environment preprocessing, enabling agents to study unfamiliar environments and create reusable artifacts without downstream task examples or model training - Built an archive guided meta agent and evaluation system using isolated Modal sandboxes and Harbor task execution with budget controls, resumable campaigns, artifact validation, and frozen downstream handoffs - Evaluated five methods on 2,396 tasks across six benchmarks and 36 environments, with meta agent variants ranking first on five benchmarks and lifting Avg@3 by up to 17.6 points while baseline inference cost was up to 5.5 times higher at matched performance	

RESEARCH EXPERIENCE

University of Maryland CLIP Group	College Park, MD
<i>Research Engineer, Advised by Prof. Mohit Iyyer</i>	Jan 2026–May 2026
- Designed REDIPO, an offline DPO data construction pipeline to recover distributional diversity in post-trained LLMs while preserving alignment, achieving up to 134% improvement in response diversity (NoveltyBench distinct-k) across 3 model families - Engineered marginal diversity scoring via embedding-based cosine similarity and quality-bounded preference pair construction to isolate diversity signal from quality thereby preventing DPO from collapsing to quality-only optimization - Fine-tuned Qwen3-4B, OLMo-3-7B, and LLaMA-3.1-8B with LoRA and DPO; improved safety (lowest HarmBench ASR on all models) while maintaining instruction-following quality (MTBench, IFEval)	
Stanford NLP Group	Palo Alto, CA
<i>Research Scientist, Advised by Prof. Diyi Yang</i>	June 2024–Feb 2025
- Built Co-Gym, an interactive platform for human-AI collaboration across travel planning, literature reviews, and data analysis - achieving 86% win rate in travel planning vs. autonomous baselines - Developed production-ready Related-Works generation system enabling dynamic interaction between researchers and LLM agents for automated literature synthesis	
CMU LTI	Pittsburgh, PA
<i>Lead ML Research Scientist, Advised by Prof. Daphne Ippolito</i>	May 2024–May 2025
- Engineered novel control framework for precise attribute control in LM generation (word count), achieving up to 58.54% improvement in constraint satisfaction - Scaled model training pipeline using LoRA and full fine-tuning of LLaMA-3-8B with Accelerate, DDP, and DeepSpeed - reducing training time by 60% - Curated 50k instruction-tuning dataset for benchmarking controllable generation for model evaluation	

Princeton NLP Group

Lead ML Research Scientist, Advised by Prof. Karthik Narasimhan

Princeton, NJ

Mar 2024–Jan 2025

- Built the **PersonaGym evaluation platform** and PersonaScore metric for **LLM persona agent evaluation** - deployed by **30+ researchers** for model assessment
- Benchmarked **10 state-of-the-art LLMs** across 200 personas with 10K evaluation questions, identifying key performance gaps
- Optimized **prompt engineering pipeline**, reducing evaluation cost by **70%** while maintaining accuracy

SELECTED PUBLICATIONS

- **[Agent]** Vinay Samuel, Varun Ursekar, Vijay S. Kalmath, Apaar Shanker, Veronica Chatrath, Yuan Xue. *Studying Without a Syllabus: Task-Agnostic Environment Preprocessing*. Under Review [Paper]
- **[LLM]** Vinay Samuel, Yapei Chang, Mohit Iyyer. *Recovering Diversity Without Losing Alignment: A DPO Recipe for Post-Trained LLMs* Under Review [Paper]
- **[Agent]** Yijia Shao, Vinay Samuel, Yucheng Jiang, John Yang, Diyi Yang. *Collaborative Gym: A Framework for Enabling and Evaluating Human-Agent Collaboration*. ICLR 2026 [Paper]
- **[LLM]** Vinay Samuel, Harshita Diddee, Yiming Zhang, Daphne Ippolito. *CIE: Controlling Language Model Text Generations Using Continuous Signals*. EMNLP 2025 (Main) [Paper]
- **[Agent]** Vinay Samuel, Henry Peng Zou, Yue Zhou, ..., Ameet Deshpande, Karthik Narasimhan, Vishvak Murahari. *PersonaGym: Evaluating Persona Agents and LLMs*. EMNLP 2025 (Findings) [Paper]
- **[LLM]** Vinay Samuel, Yue Zhou, Henry Peng Zou. *Towards Data Contamination Detection for Modern Large Language Models: Limitations, Inconsistencies, and Oracle Challenges*. COLING 2025 [Paper]
- **[Evaluation]** Yiming Zhang, Harshita Diddee, ..., Vinay Samuel, Barry Wang, Daphne Ippolito. *NoveltyBench: Evaluating Language Models for Human-like Diversity*. COLM 2025 [Paper]
- **[Multimodal LLM]** Henry Peng Zou, Vinay Samuel, Yue Zhou, Weizhi Zhang, Liancheng Fang, Zihe Song, Philip S. Yu, Cornelia Caragea. *ImplicitAVE: An Open-Source Dataset and Multimodal LLMs Benchmark for Implicit Attribute Value Extraction*. ACL 2024 (Findings) [Paper]
- **[Data Generation]** Vinay Samuel, Houda Aynaou, ..., Aman Chadha. *Can LLMs Augment Low-Resource Reading Comprehension Datasets? Opportunities and Challenges*. ACL 2024 (SRW) [Paper]

SKILLS

- **Research:** LLMs, Multimodal Learning, NLP, Agents, Semi-Supervised Learning, Retrieval Augmented Generation (RAG), Reinforcement Learning, Question Answering, LM Controllability, Evaluation, Fine-tuning, Model Parallelization, Data Preprocessing, Data Analysis, GPU, CPU, Self-Improvement, DPO, GRPO
- **Languages and Platforms:** Python, PyTorch, R, LaTeX, MATLAB, Colab, Git, SQL, C++, Triton
- **Soft Skills:** Self-Motivated, Team-Spirited, Leadership, Critical Thinking, Collaboration, Analytical, Problem Solving, Strong Communicator

COURSES

- **LLM & Agent:** Large Language Models, AI Problem Solving and Representation
- **Machine Learning:** Introduction to Machine Learning, Convex Optimization, Introduction to Deep Learning
- **Natural Language Processing:** Advanced Natural Language Processing
- **Systems:** Systems for Machine Learning